

Status of surfactants as penetration enhancers in transdermal drug delivery

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ABSTRACT

Surfactants are found in many existing therapeutic, cosmetic, and agro-chemical preparations. In recent years, surfactants have been employed to enhance the permeation rates of several drugs via transdermal route. The application of transdermal route to a wider range of drugs is limited due to significant barrier to penetration across the skin which is associated with the outermost stratum corneum layer. Surfactants have effects on the permeability characteristics of several biological membranes including skin. They have the potential to solubilize lipids within the stratum corneum. The penetration of the surfactant molecule into the lipid lamellae of the stratum corneum is strongly dependent on the partitioning behavior and solubility of surfactant. Surfactants ranging from hydrophobic agents such as oleic acid to hydrophilic sodium lauryl sulfate have been tested as permeation enhancer to improve drug delivery. This article reviews the status of surfactants as permeation enhancer in transdermal drug delivery of various drugs.

KEY WORDS: Penetration enhancer, surfactant, skin, sodium lauryl sulfate, Tween 80

Nowadays, the transdermal route has become one of the most successful and innovative focus for research in drug delivery, with around 40% of the drug candidate being under clinical evaluation related to transdermal or dermal systems. But the barrier property of skin causes difficulties for transdermal delivery of therapeutic agents.^[1] One long-standing approach to increase the range of drugs that can be effectively delivered via this route has been to use penetration enhancers, chemicals that interact with skin constituents to promote drug flux.^[2] Skin penetration enhancers are molecules which reversibly remove the barrier resistance of the stratum corneum. They allow drugs to penetrate more readily to the viable tissues and thus enter the systemic circulation.^[3] Usually, surfactants are added to formulations in order to solubilize lipophilic active ingredients, so they have potential to solubilize lipids within the stratum corneum.^[2] Surfactants induce a concentration-dependent biphasic action with respect to alteration of skin permeability. At low concentrations, surfactants increase the permeability of the skin to many substances probably because they penetrate

the skin and disrupt the skin barrier function.^[4]

Surfactants are amphipathic molecules that consist of a non-polar hydrophobic portion usually a straight or branched hydrocarbon or fluorocarbon chain containing 8-18 carbon atoms, which is attached to a hydrophilic portion. The hydrophilic portion can be nonionic, ionic or zwitterion.^[5] Surfactants are low to moderate molecular weight compounds which contain one hydrophobic part, which is readily soluble in oil but sparingly soluble or insoluble in water, and one hydrophilic (or polar) part, which is sparingly soluble or insoluble in oil but readily soluble in water.

Classification of Surfactants

Surfactants are classified according to the nature of their polar head groups^[6] i.e.

Anionic Surfactants

Surfactants in which the hydrophilic part carries a negative charge [Figure 1]. Examples of anionic surfactants include the following:

1. Carboxylates: Alkyl carboxylates-fatty acid salts ("soaps"); carboxylate fluoro surfactants.
2. Sulfates: Alkyl sulfates (RCOO) (e.g., sodium lauryl sulfate); alkyl ether sulfates (e.g., sodium laureth sulfate).

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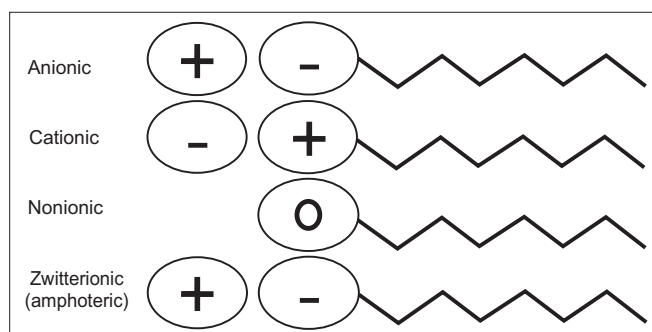


Figure 1: Schematic illustration of the various types of surfactants

3. Sulfonates: Docusates (e.g., dioctyl sodium sulfosuccinate); alkyl benzene sulfonates.
4. Phosphate esters: Alkyl aryl ether phosphates; alkyl ether phosphates

Nonionic Surfactants

Surfactants that do not ionize in aqueous solution, because their hydrophilic group is of a non-dissociable type, such as alcohol, phenol, ether, ester, or amide [Figure 1]. A large proportion of these nonionic surfactants are made hydrophilic by the presence of a polyethylene glycol chain, obtained by the polycondensation of ethylene oxide. Non ionic surfactants, i.e., surfactants with an uncharged polar head group, are probably the ones used most frequently in drug delivery applications. The critical micellization concentration for such surfactants is generally much lower than that of the corresponding charged surfactants, and partly due to this, such surfactants are generally less irritant and better tolerated than the anionic and cationic surfactants, e.g., polyoxyethylenes, poloxamer (188,407,338,184), poloxamine (304,904,908), polysorbates.

Cationic Surfactants

Cationic surfactants are dissociated in water into an amphiphilic cation and an anion, most often of the halogen type [Figure 1]. A very large proportion of this class corresponds to nitrogen compounds such as fatty amine salts and quaternary ammoniums, with one or several long chain of the alkyl type, often coming from natural fatty acids.

Zwitterionic Surfactants

Zwitterionic surfactants are less common than anionic, cationic and non ionic ones. When a single surfactant molecule exhibits both anionic and cationic dissociations it is called amphoteric or zwitterionic. The polar head group consists of a quaternary amine group and a sulfonic or carboxyl group [Figure 1]. This is the case of synthetic products like betaines or sulfobetaines and natural substances such as aminoacids and phospholipids (phosphatidylcholine, phosphatidylethanolamine).

Surfactant-Skin Interaction

Many of the properties of surfactants can be related to

their ability to concentrate at phase interfaces, leading to a reduction in interfacial tension. In biological systems the effect of surfactants are complex; particularly their effect on cell and other membranes, and this can lead to alterations in permeability characteristics.^[7] The ability of surfactants to adsorb at interfaces and bind through hydrophobic or polar interactions may lead to desirable or undesirable effects on the skin depending on the surfactant concentration, the type of exposure, the duration of contact, and the individual response. The undesirable effects of surfactants on skin have been and are still extensively studied *in vivo*, by using human volunteers, and with *in vitro* screening tests, to predict the clinical effects of an individual surfactant as well as formulations.^[8] The selection of a surfactant as a penetration enhancer should be based not only on its efficacy in enhancing skin permeation but also on its dermal toxicity (low) and its physicochemical and biological compatibility with the system's other components.^[9]

Interaction of surfactants upon contact with skin:

1. By binding to the surface proteins of the skin
2. By denaturing skin surface proteins
3. By solubilizing or disorganizing the intercellular lipids of the skin
4. By penetrating through the epidermal lipid barrier
5. By interacting with living cell.^[10]

Interaction with skin proteins

For the surfactant molecules to interact with the deeper protein - rich regions in a normal corneum, they must diffuse through the lipid region.^[11] After binding to the proteins, surfactant causes the protein to denature, leading to the swelling of stratum corneum.^[10] Solubilization of fluid lipids and abstraction of calcium or other multivalent ions to reduce corneocyte adhesion enhances the accessibility of the proteins in the lower regions of the stratum corneum.^[11] Rhein *et.al.* investigated swelling of isolated stratum corneum when exposed to various single surfactant solutions and showed that the swelling was concentration and time dependent up to the CMC before leveling off.^[12] Ionic interactions between the anionic head groups and the cationic sites of the proteins are a basic requirement to induce strong hydrophobic interactions between surfactant molecule and the protein; interactions will ultimately lead to protein denaturation.^[8] Anionic surfactants have ability to solubilize the less soluble proteins such as zein, or can themselves remain on the skin due to formation of chemical compounds with skin keratin. Anionic promote the dissolution of the proteins out of the skin, cause release of sulfhydryl groups from the sclera proteins, react upon various enzymes of skin and finally denaturize. Nonionic surfactant binds to proteins via weak hydrophobic interaction.^[13]

Interaction with the intercellular skin lipids

The protective lipid barrier of skin is composed of highly organized lipid layers located between the cells of stratum

corneum. Surfactants have to integrate into lipid bilayers to disorganize them and alter skin barrier function. Monomers of surfactant can easily reach the intercellular lipids making the effect dependent on the relative proportion of monomer in solution.^[8] Fulmer *et.al.* studied significant changes in stratum corneum lipid composition caused by sodium dodecyl sulfate [Figure 2]. The free cholesterol to cholesterol ratio was found to be increased while the quantity of total sterols remained constant. The distribution of certain ceramide species was also found to be altered.^[14] Low concentrations of surfactant may emulsify stratum corneum lipids and improve permeability; however higher concentrations promote the formation of micelles in the vehicle that trap the permeant and decrease permeability.^[15] It was found that SLS [Figure 2] enhances

penetration into the skin by increasing the fluidity of epidermal lipids. The increase in lipid fluidity below the applied site may allow sodium lauryl sulfate to diffuse in all directions including the radial path.^[16]

Interaction with living cells

Once the lipid barrier of skin has been disrupted/ weakened, monomers of surfactant can reach the living part of the epidermis and interact with keratinocytes and langerhans cells.^[8] It was found that short-term treatment of skin with sodium lauryl sulfate (0.5% v/v) disorganized the stratum corneum, induced maturation of langerhans cells, and did not result in epidermis thickening.^[17]

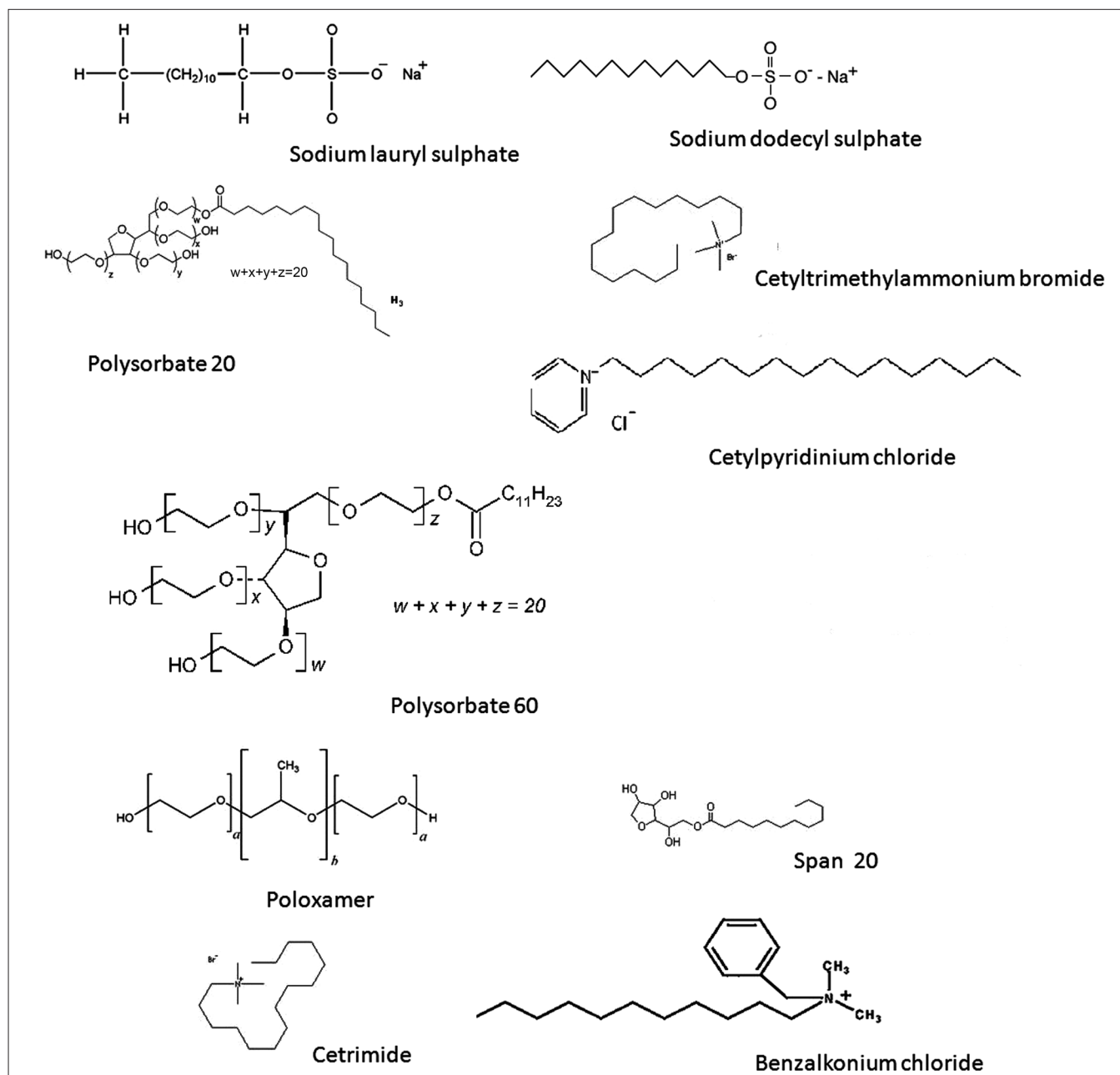


Figure 2: Structures of some of the commonly used surfactants

Factors governing the activity of surfactant as a penetration enhancer

The effect of surfactant in altering the skin barrier depends on the surfactant structure; both the hydrophobic alkyl chain and the hydrophilic ethylene oxide chain demonstrate some structure-activity behavior.^[18] In a study Walters *et al.* reported that surfactants having a linear alkyl chain greater than C8 and an ethylene oxide chain length of less than E14 caused significant increase in the flux of methyl nicotinate while those having branched or aromatic moieties in the hydrophobic portion were ineffective.^[19] In another study it was found that dimethyldialkylammoniums (double chained) with relatively shorter alkyl chains, which form either vesicles with looser molecular packing or micelles and appear to be present as surfactant monomer in higher ratios than those with longer alkyl chains, favor the interaction with skin.^[20] In many biological systems, including skin, surfactants with a similar hydrophilic group will show maximum membrane activity if they possess a decyl or dodecyl alkyl chain. Park *et.al.* reported that the enhancers containing the ethylene oxide chain length of 2-5, HLB value of 7-9 and an alkyl chain length of C16 -C18 were effective promoters of ibuprofen flux.^[21] In a study the parabolic relation between the piroxicam absorption and the polyoxyethylene length of non ionic surfactant was observed. Polyoxyethylene length from 5 to 15 was found to enhance percutaneous absorption to a greater extent.^[22] Dalvi and Zatz examined the influence of polyoxyethylene chain (POE) length of polyoxyethylenenonphenyl ether on penetration of benzocaine through hairless mouse skin from the aqueous solution system and found that the maximum flux was obtained by POE(9) followed by POE(12), POE(30), POE(50).^[12]

Sorbitan monooleate (HLB 4.3) and polyoxyethylene n-lauryl ether (HLB 12.8) were found to interact with the human epidermis in degrees which were dependent on the polarity of the surfactant.^[23] Kushla *et.al.* studied penetration of lidocaine through human epidermis by incorporating cationic surfactants of varying alkyl chain length from three classes:alkyl dimethylbenzyl ammonium halides, alkyl trimethyl ammonium halides, and alkyl pyridinium halides. Peak surfactant enhancement effects were seen at alkyl chain lengths of 12 or 14 carbons.^[24] The nature of the enhancer head group also greatly influences cutaneous barrier impairment. Tween 20 (log P_{oct}=3.72) which is more hydrophilic than Span20 (log P_{oct} = 4.26) was found to be less effective in enhancing 5-fluorouracil skin penetration. Lipophilic molecules diffuse through the stratum corneum by solubilizing in the continuous intercellular lipid phase of the stratum corneum (lipophilic pathway). Span20 was found to affect the intercellular lipids by making them more fluid and enhancing the diffusivity.^[25] SLS was found to remove detectable levels of lipids only above its critical micelle concentration (CMC). At high concentrations the surfactants removed only very small amounts of cholesterol, free fatty acid, the esters of those materials, and possibly squalene. Below the CMC, sodium lauryl sulfate bound and irritates the stratum corneum.^[26]

Mechanism of Action

Ionic surfactant

Ionic surfactant molecules in particular tend to interact well with keratin in the corneocytes, open up the dense keratin structure and make it more permeable.^[27] They are also thought to enhance transdermal absorption by disordering the lipid layer of stratum corneum.^[28]

Anionic surfactants

Anionic surfactants interact strongly with both keratin and lipids.^[29] It has been reported that anionic surfactants like sodium lauryl sulfate can penetrate and interact with the skin, producing large alterations in the barrier properties.^[30] An additional mechanism for the penetration enhancement by SLS involves the hydrophobic interaction of the SLS alkyl chain with the skin structure which leaves the end sulfate group of the surfactant exposed, creating additional sites in the membrane. This results in the development of repulsive forces that separate the protein matrix, uncoil the filaments, and expose more water binding sites, hence increasing the hydration level of skin.^[12]

Anionic materials tend to permeate relatively poorly through stratum corneum upon short-time exposure but permeation increases with application time.^[2] The alkyl sulfates can penetrate and destroy the integrity of the stratum corneum within hours of application.^[18] Anionic surfactants cause greater enhancement and damage than non ionic surfactants. Anionic has been found to increase deuterated water permeability of human epidermis *in vitro*. The surfactants such as sodium lauryl sulfate, after extended treatment, irreversibly caused protein denaturation, membrane expansion, hole formation, and the loss of water -binding capacity.^[31] Table 1 depicts the use of various anionic surfactants as enhancers in transdermal drug delivery.

Cationic surfactants

Molecules show main action on keratin fibrils of the cornified cells and result in a disrupted cell/ lipid matrix. Cationic surfactants interact with skin proteins via polar interactions and hydrophobic binding. Hydrophobic interaction between surfactant chains and the protein result in pendant ionic head groups and subsequently swelling of the stratum corneum.^[23] Cationic molecules are more destructive to skin tissues causing a greater increase in flux than anionic surfactants.^[32] Table 2 illustrates the role of cationic surfactants as penetration enhancers.

Non-ionic surfactants

Nonionic surfactants, which are a safe class of enhancers, also offer a means of enhancing drug permeation through the skin. Nokhodchi *et.al.*^[33] reported two -possible mechanisms by which the rate of transport is enhanced using nonionic surfactants.

- Firstly the surfactant may penetrate into the intercellular regions of stratum corneum, increase fluidity and eventually

Table 1: Application of anionic surfactants as penetration enhancer in transdermal drug delivery

Permeant	Surfactant as penetration enhancer	Description	References
Ketotifen	Sodium dodecylsulfate (SDS)	Significantly increased the permeation at concentrations over 1 mM. SDS induces increase in the transfer of hydrophilic ketotifen to the skin	[40]
Diazepam	Sodium lauryl sulfate (SLS)	An increase in concentration of surfactant results in an increase in the permeation rate. Highest permeation rate was obtained from the solution containing 5% W/w SLS (9.81 fold)	[35]
Lorazepam	Sodium lauryl sulfate (SLS)	Greatest increase in flux at a concentration of 5% w/w with an enhancement ratio of 11.30, compared to Tween80, CTAB and BZK in a comparative study	[33]
Propofol	Sodium dodecyl sulfate (SDS)	Shows a synergistic effect with propylene glycol and (+) menthol in delivery of propofol by acting as a membrane acting type enhancer	[41]
Naloxone	Sodium lauryl sulfate (SLS)	SLS in propylene glycol slightly increased flux but using a mineral oil vehicle showed a greater flux	[42]
Foscarnet	Sodium lauryl sulfate (SLS)	The improved efficacy of the foscarnet formulation containing SLS (5%) could be attributed to an increased penetration of the antiviral agent into the epidermis	[4]

Table 2: Application of cationic surfactants as penetration enhancer in transdermal drug delivery

Permeant	Surfactant as penetration enhancer	Description	References
Diazepam	Bezalkonium chloride (BZK)	Highest permeation rate at the concentration of 1% w/w (7.98 fold). Increasing the concentration from 1 to 5% w/w reduces the permeation rate due to interaction between micelle and drug	[35]
Lorazepam	Bezalkonium chloride (BZK)	7.66 -fold enhancement over control at 1% w/w of the surfactant	[33]
Lorazepam	Cetyltrimethylammonium bromide (CTAB)	9.82 -fold enhancement over control at 5% w/w of the surfactant. As it is more hydrophilic than bezalkonium chloride so less effective in skin permeation	[33]
Haloperidol	Cetrimide	Enhanced the permeation by increasing the thermodynamic activity which may be due to solubilization of skin lipids by micelles	[43]
Testosterone	Dodecylamine	At a concentration of 3% showed a synergic <i>in vitro</i> skin permeation enhancing -effect of N, N -diethyl -m -toluamide	[44]
Sodium nonivamide acetate	Cetylpyridinium chloride	Showed the highest iontophoretic enhancement factor and iontophoretic flux on the transdermal penetration in a comparative study with BZK and SLS. The enhancement ratio was only increased from 1.25 to 1.63 as the dose of cetylpyridinium chloride increased from 0.05% to 0.20% for four times.	[45]
Methyl nicotinate	Cetyltrimethylammonium bromide (CTAB)	Cetylpyridinium chloride can lower the resistance of the stratum corneum, such that subsequently applied iontophoresis elicits a greater transport of SNA. Exerted a greater effect on the flux of methyl nicotinate but take longer to elicit their effects. Ionic compounds can be expected to penetrate the lipophilic barrier more slowly than nonionic compounds of a similar structure	[46]

solubilize and extract lipid components.

- Secondly penetration of surfactant into the intercellular matrix followed by interaction and binding with keratin filaments may result in a disruption within the corneocyte.^[33]

In addition non ionic surfactants are able to emulsify sebum, thereby enhancing the thermodynamic coefficient of drugs and allowing it to penetrate into the cells more effectively.^[7]

Polysorbates which are a prominently safe class of surfactants offer a means of enhancing drug permeation through the skin. Polysorbate surfactants [Figure 2] demonstrated a significant increase of drug permeation up to 13 -fold when applied to hairless mouse skin.^[34] Polysorbate 80 contains the ethylene oxide and a long hydrocarbon chain and this feature imparts both lipophilic and hydrophilic characteristics to this enhancer, allowing it to partition between lipophilic mortar substance and the hydrophilic domain.^[35] Sorbitan monopalmitate, sorbitan trioleate, polyoxyl 8 stearate, polyoxyethylene 20 cetyl ether, or polyoxyethylene 2 oleyl ether 10% (w/w) significantly enhanced the percutaneous absorption of flufenamic acid and sodium salicylate when incorporated into white petrolatum USP ointment base.^[36, 37]

The effect of nonionic surfactant on the rate of the permeation of drug compound through membranes from a formulation is highly dependent on the physical state of surfactant and its concentration in the locality of the membrane.^[38] Table 3 presents the mostly used nonionic surfactants in TDDS as penetration enhancers.

Zwitterionic surfactants

Ridout *et.al.* studied the action of five zwitterionic surfactants on the barrier function of hairless mouse skin and suggested that solubilization of stratum corneum lipids may be an important mechanism of penetration enhancement. The surfactants considered for study were dodecylbetaine, hexadecylbetaine, hexadecylsulfo betaine, N, N-dimethyl -N-dodecyl amine oxide, dodecyltrimethylammonium bromide.^[39]

Regulatory Aspects of Surfactants

Safety has always been the most important requirement and the most studied when dealing with pharmaceutical drugs. In the United States, the food and drug administration (FDA)

Table 3: Application of nonionic surfactants as penetration enhancer in transdermal drug delivery

Permeant	Surfactant as penetration enhancer	Description	References
Diazepam	Tween 80	Highest permeation rate at the concentration of 1% w/w (5.68 -fold)	[35]
Tenoxicam	Polyoxyethylene alkyl ether and polyoxyethylene monoalkyl carboxylate	The flux was increased about two to four times. Enhancement was found due to the affinity of the surfactants to the stratum corneum	[47]
Diclofenac diethylamine	Sorbitan monolaurate 20	29% to 30% enhancement of skin permeation at a concentration of 0.1% w/v	[48]
Piroxicam	Polyoxyethylene-2-oleyl ether	Among the various non -ionic surfactants tested, polyoxyethylene-2-oleyl ether showed the highest enhancing effects with an enhancement factor of 2.84. The enhancers had a fluidizing effect on the lipids of the stratum corneum	[49]
Tripolidine	Polyoxyethylene-2-oleyl ether	In a comparative study of enhancers such as glycols, fatty acids and non-ionic surfactants, polyoxyethylene-2-oleyl ether showed the best enhancement at a concentration of 10%	[50]
Lidocaine	Polyoxyethylene sorbitan monoesters	Steady state flux was enhanced by surfactant at high propylene glycol concentrations	[51]
Diltiazem hydrochloride	Tween 80	Enhances the flux for approximately three times and also shortens the lag time	[52]
Amlodipine	Tween 80 + Transcutol P	20% Surfactant and 10% co surfactant exhibited highest skin permeation rate of $49.81 \pm 1.98 \mu\text{g}/\text{cm}^2/\text{hr}$	[53]
Methyl nicotinate	Brij 36 T	Brij had a smaller but immediate effect on flux in comparison to ionic surfactants of similar structure	[44]
Ibuprofen	Polyoxyethylene (POE) alkyl ethers	The permeation rate of ibuprofen was increased 0.94–5.24 times with the addition of POE alkyl ethers over the control. <i>In vitro</i> transdermal flux through excised rat skin was found in the decreasing order of POE(5) cetyl/oleyl ether ($110.24 \mu\text{g}/\text{cm}^2/\text{hr}$) > POE(2) lauryl ether ($99.91 \mu\text{g}/\text{cm}^2/\text{hr}$) > POE(2) oleyl ether ($67.46 \mu\text{g}/\text{cm}^2/\text{hr}$) > POE(10) stearyl ether ($66.19 \mu\text{g}/\text{cm}^2/\text{hr}$)	[54]
Hydrocortisone	Polysorbates	Permeation was significantly enhanced in the presence of high propylene glycol concentrations (40% or more). Synergism was attributed to enhancement of surfactant absorption by the stratum corneum leading to changes in skin barrier resistance	[55]
Benzocaine	Polyoxyethylene nonylphenol	Penetration flux was inversely related to surfactant concentration and polyoxyethylene chain length	[23]
Diclofenac sodium	Span 20, 80, 85 Tween 80	Tween 80 decreased diclofenac penetration rate due to a decrease in thermodynamic activity as a result of micellar complexation. In contrast, the more hydrophobic sorbitans enhanced penetration due to changes in the barrier properties of skin. Diclofenac flux from the gel containing Span 80 was more than 10 times larger than that from the gel with Tween 80	[56]
Atenolol	Tween 20, Brij 35, Pluronic F127	Increased the permeation of the drug in the following order: Brij 35 > Pluronic F127 > Tween 20 due to the surfactant action.	[57]
Salicylic acid	Sorbitan monopalmitate, Poloxamer 231, 182	Percutaneous absorption of salicylic acid was increased significantly when 10% w/w surfactant was incorporated.	[58]
Sodium salicylate	Sorbitan monolaurate, Poloxamer 182	Percutaneous absorption was increased significantly.	[58]

has published listings in the code of federal regulations (CFR) for GRAS substances that are generally recognized as safe.^[59]

In general, nonclinical and clinical studies are required to demonstrate the safety of a new surfactant before use. The US FDA has published a guidance document for industry on the conduct of nonclinical studies for the safety evaluation of new pharmaceutical excipients.^[60] This guidance not only provides the types of toxicity data to be used in determining whether a potential new excipient is safe, but also describes the safety evaluations for excipients proposed. The document also depicts testing strategies for pharmaceuticals proposed for short -term, intermediate, and long -term use.^[60] More importantly, this guidance highlights the importance of performing risk -benefit assessments on proposed new excipients in the drug products while establishing permissible and safe limits for the excipients. It is often possible to assess the toxicology of an excipient in a relatively efficient manner. Existing human data for some

excipients can substitute for certain nonclinical safety data. In addition, an excipient with documented prior human exposure under circumstances relevant to the proposed use may not require evaluation with a full battery of toxicology studies.^[60]

Conclusion

The effect of surfactants on the enhancement of drug permeation through skin has been well reviewed. Research in this area has proved the usefulness of surfactants as chemical penetration enhancer in the transdermal drug delivery. In many instances they have been found to be more effective than other enhancers. Focus should be on skin irritation and toxicity with a view to select from a wide range of surfactants.

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